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Labor Boundaries and Skills: The Case of Lobbyists

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Abstract. What are the determinants of in-house employment versus outsourcing in the service sector? I use detailed data on U.S. lobbying services to answer this question. I argue with a series of correlational exercises that firms tend to outsource lobbying tasks that demand a large amount of general skills, whereas they are more likely to assign firm-specific tasks to in-house lobbyists. I provide causal evidence that the need to do tasks that vary in their general skill component leads to a change in outsourcing. Using difference-in-difference estimations, I show that the 2010 British Petroleum oil spill increased the general skills needed by oil and gas firms and that, consequently, their use of lobbyists for hire increased.

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1. Introduction

Firms frequently decide whether to integrate or outsource production activities. Physical assets are often an important part of the production tasks that need to be allocated. A large stream of literature finds that different characteristics of physical assets and of the institutions surrounding them are key to explaining integration decisions. However, firms also face the same allocation decision for tasks in which physical assets are irrelevant and in which workers' human capital is the only variable influencing the benefits and costs of integration. This case is the integration decision with services industries, the sector not only with the greatest growth but also, the one with largest influence on the rest of the other industries.¹ Although integration with human assets has been recognized as more important than transacting with physical assets (Coase 1937), empirical studies on the causal link between firm boundaries and human capital needs are scarce.² In this article, I study how firms decide whether to hire in house or outsource workers from service industries.

Workers invest in skills to perform tasks that firms need to solve. Acquiring skills implies incurring fixed costs because, after the skills are acquired, the costs cannot be recovered. There are, therefore, increasing returns on the use of skills (Rosen 1983). The key insight in my article is that integration with human assets may correspond to the desire to exploit such returns. Such exploitation is possible when tasks differ in either their frequency of occurrence or the type of skills required to perform them. For instance, in the former case, returns on skills are exploited

by hiring workers when firms face a repetitive flow of tasks, whereas for infrequent tasks, the service is outsourced.

Tasks may also vary in the intensity of skills required to perform them, such as their firm and issue specificity. The former is useful for a single firm, whereas the latter is specific to a given topic but useful for a wide range of firms. I argue that in-house workers acquire skills relevant to only one firm (Becker 1964), whereas workers for hire exploit the possibility of working with several clients by acquiring expertise on issues that span a large set of firms (Abraham and Taylor 1996). Therefore, the hierarchy and the market provide different possibilities for exploiting returns on skills, and such differences determine heterogeneity in expertise: in-house service providers become firm specialists, and external ones become issue specialists. As a consequence, I predict that firms use in-house lobbyists for tasks that are frequent or require a large investment in firm-specific skills or a small investment in issue-specific skills.

These predictions apply broadly to integration with service industries. Among the many service sector demands of firms, influencing the policy-making process has become prominent in recent debates.³ Since the beginning of the lobbying industry, firms interested in lobbying the federal government have partly contracted out advocacy services to lobbyists for hire. To test the theoretical predictions, I use detailed data from the universe of firms that lobbied the U.S. federal government between 1999 and 2014.

The empirical test of these predictions requires measures of tasks' frequency, issue specificity, and

firm specificity. First, I measure firm-specific needs with RegData 2.2, a database that counts the total number of regulating words in the industry. The data measure the investment required to learn specific regulation for a firm/industry. To measure issue-specific skills, I explore the skills that firms demand when they lobby for bills. Bills are studied in congressional committees using witnesses who are experts on the issues in question. I measure the need for issue-specific tasks with the fraction of high-skill witnesses attending such committees: the greater the expertise of witnesses, the larger the investment that lobbyists need to make to learn about the issue at stake. Finally, I measure task frequency with the total number of bills studied in the congressional committees that are relevant to the firm.

I conduct estimations at both the firm-semester and firm-semester-lobbyist levels to show that firms tend to work with external rather than in-house staff when they lobby for bills with high issue-specific needs (i.e., studied by committees with a large proportion of high-skill witnesses) or when firm-specific needs are low (i.e., the number of regulating words in their industry is low). Moreover, firms facing a high frequency of advocacy tasks (i.e., bills studied by the committees that receive most bills) use in-house teams rather than external advocacy services.

An exogenous shock to the type of tasks needed by firms allows me to move from correlational to causal evidence on the impact of skill requirements on firm boundaries. I focus on tasks that differ in their issue-specific needs and test whether random variation in the need to solve these tasks leads to a change in outsourcing. I use the British Petroleum (BP) Deepwater Horizon oil spill of 2010 as an exogenous shock that increased the use of high-skill witnesses studying bills affecting the oil and gas industry. Using a difference-in-differences strategy, I show that oil and gas firms reacted to this increase in issue-specific skills by outsourcing the service more. This result not only survives a placebo exercise, but it is also robust to the test of alternative mechanisms.

The study of firm boundaries is one of the most fruitful research areas in economics and management. The literature abounds with both theoretical and empirical exercises that provide reasons for outsourcing.⁴ However, the empirical literature faces two challenges. First, it is difficult to find a data set that reports the universe of what is done in house and what is done via outsourcing, even for a single firm or industry. It is desirable to have such a universe of transactions because sampling biases may threaten the validity of exercises that use a subset of data. Second, causal evidence on channels is scarce. This sparsity in research is likely owing to the empirical challenge of finding variation orthogonal to other

independent variables. Articles that use correlation statistics face the risk of reporting false relationships among the data owing to endogeneity issues. Exceptions are given in the works of Forbes and Lederman (2009), Alfaro et al. (2016), and Gil and Ruzzier (2017), and these works provide causal evidence on reasons for integration using a large sample of data. However, none of these studies focus on integration with service industries. To my knowledge, because of both the exhaustive nature of federal lobbying reporting and the randomness of the BP oil spill, my article is the first to provide causal evidence on one of the determinants of outsourcing for service industries using the universe of both in-house and external transactions.

The remainder of the paper proceeds as follows. In Section 2, I describe the theoretical mechanism that I exploit to explain the boundary of the firm when the vertical integration is with service industries. Section 3 describes the institutional setting. Using detailed advocacy services data, Section 4 presents descriptive statistics and a series of correlational exercises that show the relationship between integration measures and types of skills needed. Section 5 presents the baseline results of the 2010 BP oil spill estimations as well as results of several robustness exercises. Section 6 discusses how I complement the existing literature and draw some conclusions.

2. A Task-based Outsourcing Model

A key distinctive feature of skills is that they are acquired at a fixed cost and therefore, provide increasing returns to their user. Both human capital and social capital in several of their forms present this feature. The aim of this article is to explore the relationship between expertise and integration patterns using these insights. The central point is that internal governance and external markets allow service providers to acquire different specialties. Firms need to perform tasks that differ in these specialties. The integration decision, therefore, depends on the type of problems that firms face. At least three considerations may affect a firm's decision to contract tasks out rather than have them dealt with in house.

First, because there are increasing returns from the use of skills and firms compensate workers for their skills, the frequency with which firms use the in-house workforce is an important variable for explaining labor boundaries. If workers are used only occasionally, it is not efficient for firms to pay the fixed cost of skill acquisition for in-house workers. Therefore, firms hire in-house workers for tasks that occur frequently and outsource tasks that occur infrequently.

Second, firms can contract out the service for tasks that require the specialized skills that in-house workers do not have. The key determinant is the degree to which economies of scale are reached with the acquired skills.

If there is not enough *market* to keep specialized external workers busy, they will acquire the same specialization levels as the in-house staff. It is the use of these skills determined by the number of clients in the external market and their needs that brings economies of scale to the learning of specialized skills. Therefore, firms outsource skill-intensive tasks when the market offers enough returns from investing in those skills.

Third, firms can develop comparative advantage by developing firm-specific skills. Because these are unique to the firm, external providers do not make investments in those skills unless there is a relational contract at place.⁵ Therefore, in the absence of such a long-term commitment, firms with high firm-specific skill requirements hire in-house workers.

Summing up, I argue that firms hire an in-house lobbyist when they face tasks that are recurrent or that require low levels of *issue expertise* or high levels of *firm expertise*. These three cases are formally presented in Section 1 of the e-companion.

3. The U.S. Federal Lobbying Market

Thanks to regulation from the U.S. government, scholars have access to detailed reports of advocacy activities. The Lobbying Disclosure Act of 1995 (LDA) requires lobbyists to register and report on their lobbying activities to the Senate Office of Public Records (SOPR). According to the act, lobbying activity is defined as contacts with officials, including background work performed to support these contacts.⁶ Two types of registrant are required to report under the LDA: external and in-house lobbyists. External lobbyists take on lobbying responsibilities for a number of different firms, and under the LDA, they are required to file a separate report for each of their client-firms. Internal lobbyists work for self-filing organizations that conduct in-house advocacy activities. Both types of registrant report good faith information every three months. Up until the end of 2007, they were required to report these estimates biannually.⁷

Each SOPR report not only contains the name of the client-firm and individual lobbyists, but also, it specifies the house(s) of Congress and federal agencies contacted as well as the issues and bills in which the client-firm was interested.⁸ Client-firms may have more than one report in a given period because they can use both internal lobbyists and one or more groups of external lobbyists. The lobbying reports data set in this study starts with the first semester of 1999 and finishes with the second semester of 2014. It contains 44,039 client-firms and 56,759 lobbyists.

The skills that lobbyists use to solve advocacy tasks can be categorized under different types. For instance,

Blanes i Vidal et al. (2012) and Bertrand et al. (2014) focus on the human capital—*knowing what*—and social capital—*knowing who*—dimensions. In this manuscript, I focus on a different categorization of these skills that encompasses both types of capital: *firm*- and *issue-specific* needs. The former relates to different issues but is specific to a given firm. The latter focuses on only one issue that is relevant across a range of firms. Therefore, these skills are general in the *Beckerian* sense, but they differ in their field of expertise.

3.1. Mapping Theory to the Lobbying Market

The U.S. federal lobbying market has two sides: firms and lobbyists. Firms need to carry out advocacy tasks, and lobbyists acquire skills for this purpose. Human capital includes knowing subjects, institutional processes, etc. Social capital includes having connections with politicians, their staff, bureaucrats, etc. An advocacy task consists of the actions that agents take to politically favor their economic situation. Such tasks usually come to firms exogenously. For instance, changes in the presidency or in the majority in Congress change the benefits of favoring specific industries or firms, and therefore, the frequency of tasks and the skills needed differ.

Tasks differ in many dimensions, and this article focuses on three of them. The first is *frequency of occurrence*. Some firms may have to lobby on many issues, such as labor regulation, tax revenues, international trade tariffs, etc. The flow of advocacy tasks that they face is, therefore, large. The business or scope of other firms, however, may be so limited that they are only affected by a few legislative proposals. For instance, a nonprofit organization specializing in defending the rights of a particular class of immigrants may have to deal with a bill affecting their interests only occasionally. This does not imply that the only relevant variation is across firms. There is also substantial variation within a firm. For instance, the political load of firms may change when they launch new products in new markets, when they suffer accidents or their industry's peers do, when there is a potential new entrant in their industry, when regulation in their vertical chain takes place, when political agendas are brought by new elected coalitions, etc.

Irrespective of the frequency of occurrence, advocacy goals may also differ in their (*issue*) *skill intensity*. Some of the bills that firms face are technical and specialized. For instance, some telecommunications companies may face regulations on geospatial technologies. Such policies usually require experts with high levels of education to explain the arguments for and against. Therefore, firms need lobbyists with high skill levels to communicate, defend positions,

and debate with these experts. Other firms may simply require extensive knowledge on how a specific request or process with the government should be made. Sometimes, the advocacy message may be easy to defend, but what firms need are connections with politicians. For both types of task, firms seek specialized expertise. This expertise is usually general in that it affects a large set of firms, but it may involve different dimensions, such as health, tax, immigration, and so on.

The last relevant dimension that this article studies is the *firm-specific requirements* of lobbying tasks. Sometimes, for lobbying to be successful, it must be specific. For instance, because firms aim to receive private benefits that depend on the specific characteristics of the firm or its products, lobbyists need to learn and fully understand the details of the firm and its industry. In this respect, Jia (2018, p. 318) states: “Successful lobbying does not typically consist of mass-produced standard services but, rather, is *customized* to closely align with the focal firm’s business operations.”

The theoretical predictions explored in this article are supported by anecdotal evidence. For instance, Drutman (2010, p. 43) cites one comment about external lobbyists (consultants): “I don’t believe most folks understand this stuff [the company-specific issues], and it’s not worth their time to get up to speed.” This implies that external staff members focus on topics that span several firms, and if firm-specific tasks are needed, in-house staff members take care of them. Drutman (2010, p. 49), citing one in-house lobbyist’s personal interview on *issue-specific* expertise, provides another spot-on example:

Most of us in the office are generalists. On new issues like energy, we didn’t know the concerns, so we need specialized talent. We’re a lean shop here, and we’re not going to hire an energy expert, so we go to the consultant who can offer a percentage of time for issue expertise. One of our consultants knows the Energy and Commerce Committee very well, so we hire them to explain what the issues are.

Referring to external advocates as *additional* lobbyists, Drutman (2010, pp. 49–50) adds,

Numerous company lobbyists said in interviews that they hire additional lobbyists for their particular issue area expertise... Like relationships, issue area expertise can be a relatively non-specific service in cases where the expertise is general enough to be of value to lots of companies.

The previous two citations support the argument about *issue-specific* expertise: lobbyists who specialize in issues are in the for-hire market and not inside

firms, and such *issue specialties* span a large set of firms. The database used in this study confirms this anecdotal evidence.

3.2. Specialization of External Lobbyists Spans Diverse Firms

In the database, external service providers have, on average, 5.9 client-firms in each semester. This is different for in-house lobbyists who, on average, work for only one firm in a given semester. If we look at the career of lobbyists, in-house lobbyists tend to work for 1.27 firms, whereas external lobbyists tend to work for 13.15 clients. This is consistent with the logic in which the external lobbyists make larger human capital investments that are general to a set of firms because the market provides the possibility of working for multiple clients.

Lobbyists also differ in their working experience. Some of these intermediaries are revolving door lobbyists (i.e., because they previously worked in federal departments and Congress). This working experience provides lobbyists with skills on issues rather than knowledge about specific firms. The reason is that government’s regulation studies topics that span several firms and industries. For instance, health issues were lobbied by about 2,000 firms.⁹ Most of these revolving door lobbyists work in the for-hire market. For instance, in my sample, 90% of ex-congressmen lobbyists work in the external market. These fractions are similar when we look at different working experience, such as staff in Congress or the White House.

Firms use lobbyists with different specialties to affect policy making in different spheres. They try to influence federal departments, executive agencies, and legislative agencies, such as the House of Representatives and the Senate. Most of the U.S. federal-level lobbying action takes place in these two chambers, which are in charge of proposing, studying, and approving bills. Bills are legislative proposals that can be introduced at any time while the Congress is in session by any member of either house. After introduction, the bill is referred to the appropriate committee or committees based on the committees’ jurisdiction, which is defined by congressional rules.¹⁰ Lobbyists make connections with politicians working in congressional committees. Those committees study *issues*, not *firms*. For instance, the U.S. Senate Committee on Health, Education, Labor and Pensions has a large range of jurisdictional areas, affecting a large set of firms. One of the topics that this committee studies is biomedical research, which affects firms primarily involved in physical research, educational establishments conducting research, pharmaceutical manufacturing, and drug wholesalers. This same committee studies other diverse topics, such as health, labor,

student loans, private pension plans, and the American Red Cross. Other committees with narrower sets of topics also affect a large range of firms.

In the next section, I present the data and empirical strategy used as well as the results of different fixed effect estimations supporting the main theoretical predictions proposed.

4. Firm Fixed Effect Estimations

4.1. Issue-specific Measures

Firms lobby for bills, and these are studied in committees that consider whether to send them for further action. The initial stage of this study process consists of public hearings at which committee members invite witnesses with the purpose of gathering relevant information. Witnesses are either specialists on the topic or people affected by the matter. They represent different views on the topic and may come from different backgrounds, such as government, academia, or business. Committees that manage more technical subjects require witnesses with more experience or higher education levels.¹¹

Focusing on the period 1999 to 2014, the Senate conducted a total of 8,181 hearings, and the House of Representatives conducted a total of 12,269 hearings. An average committee conducts 17.3 hearings in one semester. In the Senate, the committees conducting the largest average number of hearings were *appropriations*; *homeland security and governmental affairs*; and *commerce, science, and transportation*. For the lower chamber, the *government reform*, *foreign affairs*, and *commerce* committees held the largest number of

hearings, conducting, on average, 40 hearings a semester in each of them.

To complement the *SOPR* database that contains information on firms and lobbyists and to construct a measure of the skills required for *issue-specific* needs in lobbying activities, I use information from the *U.S. Government Publishing Office* (GPO). The GPO releases all of the transcribed congressional hearings on its web page. The main items of information on these hearings are committee in which the hearing was conducted, date, bills studied, and information on the witnesses. This information contains the name and title or occupation of all of the witnesses in all of the reported congressional hearings since 1999. I classified all of the occupations in three categories: (1) high-skill occupations (27%), (2) low-skill occupations (53%), and (3) unclassified occupations (20%). The first group consists of witnesses who have in their occupation title words such as PhD, professor, or senior manager. Examples of low-skill occupations are farm owners, farm producers, assistant secretaries, and citizens with an interest in the issue. Finally, *unclassified occupations* are cases in which the hearing gives a name but does not report the specific occupation of the person.¹² In this paper, I measure the issue-specific needs with the fraction of high-skill witnesses over all of the classified witnesses (high and low skill).¹³

To give a better idea on the variation of the data, Table 1 shows the four committees with the highest and lowest fractions of high-skill witnesses. In total, I report four committees for each congressional chamber. Committees that study issues requiring a higher level of skill requirements, such as *science*, *space*, and

Table 1. Statistics of the Committees and Regulating Words Across Industries.

Committees	Fraction	Regulating words	K
Least			
Homeland security, S	8.3	Support activities for forestry (1,153)	1
Rules & administration, S	9.5	Textile furnishings mills (3,141)	1
Foreign affairs, H	12.9	Household appliances & electrical and electronic goods merchant wholesalers (4,236)	1.2
Veterans' affairs, H	13.5	Alumina & aluminum production and processing (3,313)	1.8
Most			
Finance, S	18.1	Depository credit intermediation (5,221)	1,518.4
Energy and commerce, H	21.3	Nondepository credit intermediation (5,222)	1,557.9
Commerce, science, and transportation, S	25.4	Electric power generation, transmission, & distribution (2,211)	1,675.4
Science, space, and technology, H	32.5	Petroleum & coal products manufacturing (3,241)	2,547.9

Notes. On the left-hand side, an observation is a committee. The table calculates the fraction of high-knowledge witnesses for each committee across all of the semesters between 1999 and 2014. This fraction is reported in the second column, Fraction. The table presents the four committees with the largest fraction and the four committees with lowest fraction of high-knowledge witnesses. In the first column, S (H) means that the committee belongs to the Senate (House of Representatives). On the right-hand side, an observation is a four-digit industry. The table shows the most (lower section) and least (upper section) four-digit industries by the total number of regulating words according to RegData 2.2. The table uses data for 2014. The fourth column, K, reports the total number of regulating words in 1,000 units. Each industry reports its corresponding *North American Industry Classification System* four-digit industry code.

technology from the House and *commerce, science, and transport* from the Senate, have the largest fractions of high-skill witnesses. This provides suggestive evidence that the use of witnesses to classify the skill intensity of the committees gives intuitive results.

4.2. Firm- and Industry-specific Measures

An SOPR report contains basic information on the name of the firm, lobbyists, and lobbying activities. Ideally, I would like to construct a skill measure at firm level, but data availability limits this possibility. Instead, I construct a measure that varies at industry level. The empirical arguments arguing that external lobbyists specialize in issues and not industries and that these issues span a large set of industries are still valid. Firms lobbying at the U.S. federal level belong to a large set of industries. For instance, according to the SOPR data, over 100 three-digit North American Industry Classification System (NAICS) code industries actively lobby on activities registered as health issues. This is also true for not heavily lobbied issues, such as advertising because, for instance in 2014, firms lobbying on this issue belong to NAICS codes as diverse as 5182, 5221, 4452, 3399, and 5416. In the database, external service providers have, on average, clients in each semester belonging to 3.4 three-digit code industries. In contrast, in-house lobbyists work for only one industry in a given semester.

All industries have their own specificity. Some are more regulated than others. Lobbyists need to understand the political and regulatory environment of the industries of the firms for which they want to advocate. Without this knowledge, it is difficult to detect opportunities and challenges for firms. A more regulated industry requires making larger skill investments, and therefore, the more regulated and the more specific it is, the less likely it is that the external market can offer the skills needed because external lobbyists will face a thinner market to obtain returns from these investments.

To construct a measure with *industry-level* variation, I use the RegData 2.2 of Al-Ubaydli and McLaughlin (2017). This database uses text analysis and machine learning algorithms to create regulation intensity measures at the *industry-year* level based on the *Code of Federal Regulations* (CFR).¹⁴ The CFR is an *annual* codification of rules drawn up by executive and federal government agencies.¹⁵ RegData classifies industries at the NAICS four-digit code levels for the period 1999 to 2014. In order to match the RegData information with the lobbying reports, I detect the industry of the firms using ORBIS, COMPUSTAT, SIC-CODE, and other web sources, such as firms' web pages. I use the total number of regulating words for each industry-year combination as a measure of *firm-specific* needs.

Table 1 shows the four industries with the largest and smallest total numbers of regulating words in 2014. Similar rankings are obtained for other periods, showing a relative stability in the differences of regulatory environments across industries.¹⁷ Table 1 shows that companies engaged in *textile furnishings mills* as well as establishments *supporting activities for forestry* belong to the least regulated industries. Anecdotal evidence shows that their advocacy strategies tend to rely heavily on in-house staff.

4.2.1. RegData as a Measure of Industry-specific Skills.

The literature on work displacement has shown strong evidence on the existence of *industry-specific* skills. These articles show that workers who switch industries following displacement have significantly larger earnings losses than workers who remain in the same industry after displacement. Using the National Longitudinal Survey of Youth and the Panel Study of Income Dynamics, Parent (2000) shows evidence of both *firm-* and *industry-specific* skills.

The intuition used by this literature to measure industry specificity can be taken one step further. If the earnings losses from switching industries proxy for *industry-specific* skills, the level of the losses may proxy for the level of this specificity. That is, the industries in which workers suffer most if they leave will be industries with a higher level of *industry-specific* skills.

Based on this intuition, I conduct validation exercises for the measure of *industry specificity* based on RegData. I use *industry specificity* measures from the displaced workers' literature and the trade literature. Both of these sources provide a noncomprehensive cross-section of empirical measures for *industry specificity*. Section 3 in the e-companion provides the results from these exercises, concluding that, given the data limitations faced, RegData provides a reasonable *industry specificity* measure.

4.3. Empirical Strategy

I run the following two sets of *fixed effects* estimations:

$$VI_{ijt} = \beta_j + \gamma_t + \eta X + \varepsilon_{ijt}$$

$$fr_{jnt} = \beta_j + \gamma_t + \eta X + \varepsilon_{jnt},$$

where i indicates a report, j indicates a firm, n indicates an industry, and t indicates a time period, which can be either a semester or a year. The unit of observation in the first regression is a lobbying report, whereas it is a firm-period in the second regression. At the report level, VI_{ijt} is a dummy variable that takes the value of one when the report is conducted

internally and zero otherwise. At the firm level, I use fr_{jnt} to denote the fraction of in-house lobbying reports over the total number of reports that the firm has in period t . I control for both firm (β_j) and time fixed effects (γ_t). I include firm dummies to control for mean differences in the dependent variable across firms and time dummies γ_t to control for growth in the dependent variable that is common to all firms. Intuitively, I include the former dummies because there may be some omitted *firm-level* characteristics, such as labor union status, geographical variation (i.e., location of the firm's divisions), or relationship with politicians and the federal government, all of which may confound the outsourcing decision.¹⁷ I use *time fixed* effects because there may be some time-varying changes in the dependent variable as a result of the financial crisis or changes in market uncertainty among other reasons.¹⁸ For most of these exercises, I cluster the standard errors at the firm level. I also include *industry-year fixed* effects to control for time variation within an industry because the level of integration and regulation may evolve differently across industries over time (see, for instance, the BP case in Section 5). I cluster the standard errors at the industry level when I include industry fixed effects or at the *industry-year* level for the latter set of controls. The key independent variable in these estimations, X , represents *issue-specific* needs, *firm-specific* needs, or frequency.

1. **Issue-specific needs.** I measure this variable using the information on the committees. Ideally, because the lobbying reports provide information on the bills lobbied, I would like to consider *issue specificity* measures at the bill level. However, this is not feasible. First, not all of the bills are studied in congressional

hearings, and some of these meetings study a group of bills (rather than one bill) spanning a common topic. Second, because lobbyists can choose the specific bills that they lobby for, the inclusion of a measure at the bill level may bias the estimates owing to double causality. In consequence, I aggregate the fraction of high-skill witnesses at the committee-semester level. I then match these measures with each of the lobbying reports according to the information on which committees the bills were assigned to.¹⁹ This ensures that all of the bills receive *issue-specific* measures, and more importantly, I avoid obvious double-causality problems in my estimations.

2. **Firm-specific needs.** I use the total number of regulating words for each industry-year combination.

3. **Frequency.** Using the variation of the congressional committees, I use the total number of bills studied in the committees that study bills affecting the firm. Because a given lobbying report may cover more than one bill, the frequency measure is the weighted average over all of the frequencies of the committees.²⁰

4.4. Descriptive Statistics

Table 2 shows the mean, standard deviation, and total number of observations for the main variables in my analysis: VI , fr , and measures of *issue specificity*, *firm specificity*, and frequency. The unit of observation used to construct this table is the *firm-semester* for all of the variables except VI , which is constructed at the report level. The variation in the independent variables is at the *firm-semester* level for the measures based on the congressional committees and at the *industry-year* level for the measures based on RegData. The number of observations for the *issue-specific* need

Table 2. Descriptive Statistics

	Mean	Standard deviation		Observations
VI	0.135	Total	0.342	409,515
fr	0.180	Total	0.367	176,143
	0.180	Between	0.260	176,143
	0.180	Within	0.076	176,143
Issue-specific S. (committees)	24.65	Total	29.17	36,477
	24.65	Between	17.66	36,477
	24.65	Within	18.91	36,477
Firm-specific S. (/100)	3,668.27	Total	7,365.37	176,143
	3,668.27	Between	7,817.72	176,143
	3,668.27	Within	730.37	176,143
Frequency	359.38	Total	398.67	36,477
	359.38	Between	268.29	36,477
	359.38	Within	279.54	36,477

Notes. The unit of observation used to construct this table is the firm-semester for all of the variables except VI , which is constructed at the transaction level. The table shows the mean, standard deviation (total, between, and within), and total number of observations for the main variables: VI , fr , and measures of *issue-specific* and *firm-specific* skills as well as frequency. The number of observations for the measures of frequency and the *issue-specific* skills is smaller than for the other variables because only a fraction of firms lobby for bills.

and for frequency is smaller than for the other variables because only a fraction of firms lobbies for bills. On average, 50% of the firms report advocating for bills.

The table shows that, across all of the reports, 13.5% were filed by in-house lobbyists. Focusing on the variation within firms, on average, 18% of their reports were in house. An average *firm-semester* combination lobbies for bills that are assigned to committees that receive 360 bills, and when congressional hearings occur, 24% of witnesses are classified as high skill. The data set shows substantial variation. For instance, for the integration variable *fr*, there is larger variation across firms than for a single firm across time. This provides some evidence on inertia in lobbying activities.²¹ This same pattern is also present in the *firm specificity* variable because most of its variance is explained by differences across industries. However, it is important to note that there is also enough within-industry variation. Finally, all of the variables present substantial within-firm variation. For instance, for *fr*, the within-firm standard deviation is 42% of the mean value. Although this variation is lower for *firm specificity* (20%), it is larger for the variables based on the committees. For both of these variables, the within-firm standard deviation is about 78% of the mean value. For the sake of interpretation, I normalize

all of the independent variables by their standard deviation in the following econometric exercises.

4.5. Results

4.5.1. Issue-specific Needs. Table 3 shows the results for this measure based on the witnesses in committees dealing with the bills for which the firms lobby. This table is organized as follows: odd columns control for both *firm* and *semester fixed* effects, the second and sixth columns control for *industry · year fixed* effects, and finally, the fourth column controls for both *firm* and *year fixed* effects. The results of this table are divided into two sections: the upper section shows the results at the report level, and the lower section shows the estimations at the firm level. Other tables from this section are organized similarly.

As the theoretical section predicts, the greater the *issue-specific need*, the lower the level of vertical integration. In this table, all of the coefficients are significant and negative at least at the 10% level. The results imply that an increase of one standard deviation in the fraction of high-skill witnesses is associated with a decrease in the probability that advocacy activities will be conducted by in-house staff of between 3.3 and 3.7 percentage points (lower section). Given that the mean value of the dependent variable *fr*

Table 3. Relationship Between Issue-specific Skills, Firm-specific Skills, Frequency, and Vertical Integration

Report level: Dependent variable <i>VI</i>						
<i>Issue-specific skills</i>	−0.0271** (0.0121)	−0.0224* (0.0124)				
<i>Firm-specific skills</i>			0.0046* (0.0027)	0.0045* (0.0025)		
<i>Frequency</i>					0.0162*** (0.0032)	0.0112** (0.0057)
Observations	54,472	54,472	297,916	297,916	54,472	54,472
Adjusted R ²	0.662	0.846	0.769	0.769	0.562	0.852
Firm level: Dependent variable <i>fr</i>						
<i>Issue-specific skills</i>	−0.0377** (0.0186)	−0.0330* (0.0191)				
<i>Firm-specific skills</i>			0.0037* (0.0021)	0.0037* (0.0021)		
<i>Frequency</i>					0.0192*** (0.0032)	0.0144** (0.0066)
Observations	36,467	36,467	176,143	176,143	36,467	36,467
Adjusted R ²	0.743	0.945	0.956	0.956	0.680	0.979
Fixed effects						
<i>Firm</i>	Y		Y	Y	Y	
<i>Semester</i>	Y		Y		Y	
<i>Industry × Year</i>		Y				Y
<i>Year</i>				Y		

Notes. An observation is a lobbying report (upper section) or a firm-semester combination (lower section). The table shows the coefficients obtained when the independent variables are frequency, issue (fraction of high-knowledge witnesses), and firm-specific skill measures (number of regulating words). Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the firm level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

*Significance at 10%; **significance at 5%; ***significance at 1%.

is 18%, the estimates represent a decrease of about 21% in the dependent variable. Similar results emerge when I consider *report-level* (upper section) estimations.

4.5.2. Firm-specific Needs. Table 3 shows the relationship between vertical integration and firm specificity when the latter is measured with the number of regulating words of the industry that the firm belongs to. The table shows that all of the coefficients are statistically significant and positive as the theoretical section predicts. In particular, a one-standard deviation increase in the total number of words regulating an industry is associated with 0.45% of the probability of vertical integration (upper section) and 0.37% of the fraction of in-house reports (lower section). Considering the mean values of the dependent variables, these effects represent decreases in the probability of integration of three percentage points (upper section) and 2% (lower section) in the fraction of internal reports.

4.5.3. Frequency. Table 3 shows the results when the frequency of advocacy tasks is measured with the average number of bills introduced in all of the committees studying bills for which a given firm lobbies.

In this table, all of the coefficients of interest are statistically significant, and their signs are consistent with the theoretical predictions. The results imply that an increase of one standard deviation in the number of bills introduced is associated with an increase in the probability of vertical integration of between 1.1 and 1.6 percentage points (upper section). Similarly, an additional standard deviation in the number of bills introduced is associated with an increase in the fraction of in-house reports of between 1.4% and 1.9% (lower section). These effects are economically relevant. For instance, in the latter case, a one-standard deviation increase in the frequency variable is associated with an increase in the fraction of reports made by in-house lobbyists of between 7% and 11%.

4.5.4. Frequency and Skill Needs. An additional exercise is to estimate the relationship of firm- and issue-specific needs controlling for the changes in frequency of tasks faced by firms. The rationale behind this exercise is that changes in the composition of witnesses or regulating words may be negatively correlated with the number of bills affecting the firms.

Table 4. Relationship Between Issue-specific, Firm-specific Skills and Vertical Integration Controlling by Frequency of Tasks

Report level: Dependent variable <i>VI</i>				
<i>Issue-specific skills</i>	−0.0250** (0.0116)	−0.0221* (0.012)		
<i>Firm-specific skills</i>			0.0049** (0.0023)	0.0046** (0.0022)
<i>Frequency</i>	0.0172*** (0.0030)	0.0114* (0.0054)	0.0171*** (0.0031)	0.0183*** (0.0034)
Observations	54,472	54,472	54,472	54,472
Adjusted <i>R</i> ²	0.691	0.863	0.874	0.817
Firm level: Dependent variable <i>fr</i>				
<i>Issue-specific skills</i>	−0.0370** (0.0181)	−0.0327* (0.0198)		
<i>Firm-specific skills</i>			0.0039** (0.0018)	0.0041** (0.0019)
<i>Frequency</i>	0.0195*** (0.0031)	0.0146** (0.0062)	0.0193*** (0.0030)	0.0196*** (0.0027)
Observations	36,467	36,467	36,467	36,467
Adjusted <i>R</i> ²	0.782	0.981	0.980	0.967
Fixed effects				
<i>Firm</i>	Y		Y	Y
<i>Year</i>				Y
<i>Semester</i>	Y		Y	
<i>Industry × Year</i>		Y		

Notes. An observation is a lobbying report (upper section) or a firm-semester combination (lower section). The table shows the coefficients obtained when the independent variables are frequency, issue (fraction of high-knowledge witnesses), and firm-specific skill measures (number of regulating words). Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the firm level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

*Significance at 10%; **significance at 5%; ***significance at 1%.

This would imply that the results can be explained simply by changes in the political workload, and as a consequence, the relationship of skills measures on vertical integration found previously simply reflects the relationship between skills requirements and frequency of advocacy tasks.

Table 4 presents results at odds with this possibility. All of the coefficients representing skill requirements are significant and have the same sign reported in the tables in which each measure of skill needs was calculated separately. The coefficient of the skill requirements represents the relationship of this variable with the measures of vertical integration holding the workload faced by these firms constant. However, the coefficients of firm- and issue-specific skills slightly increase when the frequency variable is included.²² This is consistent with a negative but slight correlation between the frequency variable and the measures of skill requirements. The important takeaway is that the results on the relationship between skills and integration survive in a robust way, even controlling for changes in the workload of advocacy activities.

4.5.5. Effects on Number of Lobbyists. As dependent variables, all of the previous estimations take the fraction of reports made internally or the dummy of in-house tasks. This does not tell us much about the use of in-house and external lobbyists given that a rise in the proportion of in-house reports could be explained by an increase in the use of in-house lobbyists, a decrease in the use of external lobbyists, or both. Table 5 presents estimates that support the idea that firms modify their use of *both* types of lobbyist when the skills needed change or the frequency of tasks changes. The estimates for the three key independent variables for the total number of in-house lobbyists are given in the upper section of the table, and the estimates for lobbyists for hire are given in the lower section of the table. Firms tend to hire fewer in-house lobbyists and tend to increase the number of for-hire lobbyists when the issue-skill requirements for advocacy increase. These patterns are the reverse to increases in either the firm-specific skill or the frequency measures. The results are statistically significant, and as in Table 4, the effect of both skill measures on the decision regarding both types of

Table 5. Relationship Between Issue-specific Skills, Firm-specific Skills, Frequency, and Number of Lobbyists

Firm level: Dependent variable <i>N. Internal Lobbyists</i>				
<i>Issue-specific skills</i>	−0.421*** (0.142)	−0.418** (0.174)		
<i>Firm-specific skills</i>			0.054* (0.028)	0.065** (0.029)
<i>Frequency</i>	0.126*** (0.046)	0.118* (0.064)	0.134** (0.053)	0.132** (0.061)
Observations	36,467	36,467	36,467	36,467
Adjusted <i>R</i> ²	0.623	0.855	0.860	0.798
Firm level: Dependent variable <i>N. External Lobbyists</i>				
<i>Issue-specific skills</i>	0.125** (0.049)	0.118** (0.047)		
<i>Firm-specific skills</i>			−0.018* (0.010)	−0.045* (0.023)
<i>Frequency</i>	−0.145** (0.058)	−0.158* (0.087)	−0.124* (0.065)	−0.165** (0.073)
Observations	36,467	36,467	36,467	36,467
Adjusted <i>R</i> ²	0.741	0.936	0.923	0.942
Fixed effects				
<i>Firm</i>	Y		Y	Y
<i>Year</i>				Y
<i>Semester</i>	Y		Y	
<i>Industry × Year</i>		Y		

Notes. An observation is a firm-semester combination. The table shows the coefficients obtained when the independent variables are frequency, issue (fraction of high-knowledge witnesses), and firm-specific skill measures (number of regulating words). The dependent variable in the upper section is the total number of in-house lobbyists hired for a firm in a given period. The dependent variable in the lower section is the total number of external lobbyists. Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the firm level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

*Significance at 10%; **significance at 5%; ***significance at 1%.

lobbyist holds, even when changes in the political workload are controlled for.

The previous estimations use firm- or report-level variation. In the next subsection, I argue that the relationship between specialization and integration holds when the unit of analysis is at the lobbyist level.

4.6. Lobbyists That Cross the Boundary

Using lobbyists' issue assignments from the SOPR data, I construct a *Herfindahl* concentration index (HHI) for lobbyists, and following Bertrand et al. (2014), I categorize them into two possible corner solutions: generalists and specialists.²³ A lobbyist is a generalist if more than 25% of her assignments are never on the same issue, whereas she is a specialist if at least 25% of her assignments are on the same issue.²⁴ The data show that, among the ones classified as generalists, 78% are in-house lobbyists, and among the external lobbyists, 79.1% are classified as specialists. Figure 1 shows that in-house lobbyists work for more issues and correspondingly, have a lower HHI.

To study the relationship between specialization and integration, I regress

$$s_{it} = \beta_i + \gamma_t + \text{Tenure}_{it} + v_{it} + \varepsilon_{it},$$

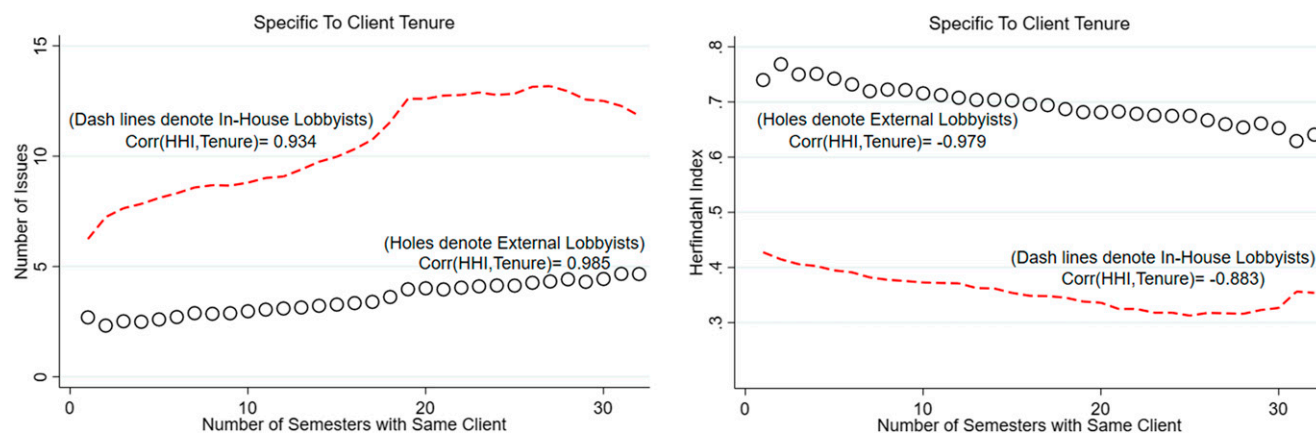
where s_{it} is one of the following five measures of specialization: (1) *Herfindahl index* of the lobbyist i in semester t , HHI_{it} ; (2) Specialist_{it} takes a value of one at period t if lobbyist i is considered specialist; (3) Generalist_{it} takes a value of one if lobbyist i is considered generalist at period t ; (4) Issues_{it} is the total number of lobbied issues for the focal lobbyist at period t ; and (5) Bills_{it} represents the total number of advocated bills at period t for lobbyist i . Tenure_{it} is the accumulated number of semesters that the focal

lobbyist has worked in the U.S. federal lobbying industry. The key independent variable is v_{it} , which takes the value of one when the lobbyist is working in house and zero otherwise.²⁵ Given that the estimations control for lobbyists' fixed effects, the variation in the independent variables is given by the lobbyists who work in house for some periods and externally for other periods. In my sample, 56% of lobbyists have always been in house, 37% have always been external, and the remaining 7% are lobbyists who switch.

Table 6 shows the results of this exercise. Controlling for the tenure of the lobbyists, external lobbyists who become in-house advocates decrease their HHI index and are less likely to be specialists. There is no statistical evidence that lobbyist are more likely to be generalists after they become in house. However, the coefficient of integration status on the dummy of being a generalist is positive, denoting an increase in the probability of working on several issues after the lobbyist is hired in house. Confirming this intuition, Table 6 also shows that advocates who change their contractual arrangement from external to in house lobby for both more issues and more bills.

The results presented provide empirical evidence that the insights proposed in Section 2 are valid in the U.S. lobbying industry. The coefficients of key independent variables are significant and have the expected sign. Their significance survives different types of *fixed effect* controls for estimations at the report, firm, and lobbyist levels. The tables present empirical correlations, and no causality can be inferred from them. To advance in this direction, in the following section, I consider an exogenous change that modified the *issue-specific* skill requirements for a subset of firms from my sample.

Figure 1. (Color online) Number of Issues Lobbied and HHI index by Integration Status and Client-Specific Tenure



Notes. The figure presents the relationship between the number of issues lobbied (and HHI index) and tenure of the same client. The lobbyists are classified according to their integration status (in house and external). Dashed lines denote in-house lobbyists. The plots containing holes denote external lobbyists. The x axis represents the number of semesters that lobbyists have worked for the same client. For instance, the number 10 on the x axis of the left panel presents the average number of issues worked for in-house and external lobbyists in their 10th semester. Similar interpretation follows for other years. All of the reported correlations are significant at 5%.

Table 6. Lobbyists Who Cross the Boundary of the Firm

Variables	HHI_{it}	$Specialist_{it}$	$Generalist_{it}$	$Issues_{it}$	$Bills_{it}$
v_{it}	−0.028*** (0.008)	−0.016* (0.009)	0.009 (0.010)	0.031** (0.015)	0.023** (0.009)
$Tenure_{it}$	−0.003*** (0.000)	−0.003*** (0.001)	0.002*** (0.000)	0.027* (0.014)	0.002*** (0.000)
Observations	368,422	368,422	368,422	368,422	294,568
Adjusted R^2	0.75	0.68	0.65	0.77	0.73
Fixed effects					
<i>Lobbyist</i>	Y	Y	Y	Y	Y
<i>Semester</i>	Y	Y	Y	Y	Y

Notes. An observation is a lobbyist-semester combination. There are five independent variables: (1) Herfindahl index (HHI), (2) probability of being a specialist ($Specialist$), (3) probability of being a generalist ($Generalist$), (4) number of issues lobbied ($Issues$), and (5) number of bills lobbied ($Bills$). The table includes $Tenure_{it}$, which represents the accumulated number of semesters that the focal lobbyists have worked in the U.S. federal lobbying industry. v_{it} is a dummy that takes the value of one when the lobbyist is working as an in-house lobbyist and zero otherwise. The last column contains fewer observations because lobbyists do not work for bills every semester. Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the lobbyist level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

*Significance at 10%; **significance at 5%; ***significance at 1%.

5. The 2010 BP Oil Spill

On April 20, 2010, high-pressure methane gas from the Deepwater Horizon oil well rose into the drilling rig, where it ignited and exploded. This explosion led to the burning and sinking of the Deepwater Horizon drilling rig. It was followed by a massive offshore oil spill in the Gulf of Mexico that was considered one of the largest environmental disasters in U.S. history.²⁶ The executive branch of the government and the Congress reacted to this event.

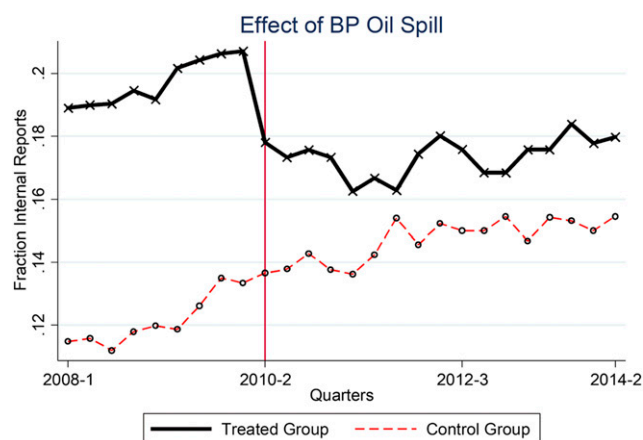
In June 2010, U.S. President Barack Obama created a nonpartisan national commission to provide a deeper understanding of the BP oil spill. The commission investigated the causes of the spill and released a final report in January 2011, concluding that not only could the well blowout have been prevented but also, that it was an example of the failure of risk-prevention practices within the entire oil extraction industry. The commission agreed that deep-water exploration has intrinsic risks but that it is the responsibility of both the industry and the regulatory agencies to restructure the way that business is being done to improve safety throughout the industry. Thus, the report blamed both oil extracting firms and regulators: the former for being irresponsible in their safety practices as confirmed by the evidence on 79 well control accidents between 1996 and 2001, many because of negligence by oil and gas extracting firms, and the latter for lagging behind on regulating the real risks associated with deep-water drilling. Therefore, this report increased the pressure to improve regulatory oversight of the industry.

In the months following the spill, there were dozens of protests.²⁷ Interviews with protesters showed that

the aim of the demonstrations was not only to damage BP's reputation but also, to show public discontent with the operation of the whole extraction industry without adequate safeguards (Wheaton 2010). The U.S. Congress reacted to the reception of the spill by the public and the executive by proposing new regulations. For instance, right after the spill, in May 2010, Roy Blunt introduced bill H.R. 5356 with the purpose of raising the cap on liability for economic damages resulting from such a spill. This higher cap led to an increase in the expected cost of oil spills to firms at risk because they would bear a greater responsibility for damages.²⁸ In the next section, I show that the BP oil spill increased the *issue-specific* needs of lobbying activities for firms belonging to the oil extracting industry. I then show that firms from this industry responded to this increase by outsourcing more.

5.1. Empirical Strategy and Baseline Results

BP agreed to pay £18.7 billion for damages caused, and as a consequence, the company sold off \$38 billion in assets from 2010 to 2012.²⁹ This change in the internal structure of the firm may confound its decision to hire in-house lobbyists. To conduct difference-in-difference estimations, I need both treated and control groups. For the former group, I use all of the firms conducting lobbying activities that belong to the same subindustry as BP (2007 NAICS code 211111) *excluding* BP. This classification gives me 52 firms. Examples are Chevron, Exxon Mobil, Phillips 66, Shell, and Devon. As a control group, I include all of the firms that belong to the oil and gas industry, excluding the 2007

Figure 2. (Color online) BP Oil Spill and Fraction of Internal Reports for Firms in the Oil Industry.

Notes. An observation is a group-quarter. The treated group includes all of the firms conducting lobbying activities that belong to 2007 NAICS codes 211111, excluding BP. The control group includes all of the firms that belong to the oil and gas sector, excluding 211111 and the ones that have links with oil and gas extraction (i.e., 213111 and 213112).

NAICS code 211111. I also exclude code 213111 (*drilling oil and gas wells*) and code 213112 (*support activities for oil and gas operations*) because they are linked to the affected industry. Examples of industries in this control group are *basic chemical manufacturing* (NAICS 3251), *petroleum lubricating oil and grease manufacturing* (NAICS 324191), and *all other petroleum and coal products manufacturing* (NAICS 324199). The group is composed of 218 firms.

Figure 2 shows the time series patterns of in-house reports for the control and treated groups. The graph shows two main patterns. First, the dependent variables for both groups have similar trends before the spill. This is confirmed by statistical exercises, such as leads and lags estimations and *t*-test results. Figure 3 in the e-companion presents the coefficient estimates from the leads and lags exercise. Second, for the control group, there seems to be no change in the increasing trend around the second quarter of 2010, during which the BP oil spill took place. However, the fraction of internal reports for the treated group drops significantly in the quarter in which the BP oil spill occurred. The trend then flattens, and it does not show any tendency toward returning to the levels prior to the spill.

To estimate the effect of the spill, I regress integration measures on firm and quarter fixed effects and an interaction of the postspill period with an indicator for whether the firm belongs to the same subindustry as BP:

$$v_{jt} = \beta_j + \gamma_t + \delta(T_j \cdot P_t) + \varepsilon_{jt},$$

where v_{jt} is a measure of vertical integration made by the j th firm at quarter t . I consider two measures for

this variable: fraction of in-house reports and fraction of lobbyists working in house. β_j (γ_t) represents firm-level (quarter-level) fixed effects. The variable T_j takes a value of one if firm j belongs to the same subindustry as BP (NAICS 211111) and zero otherwise. The variable P_t takes a value of one for all of the quarters since the second quarter of 2010 until the end of the sample and 0 for periods before the oil spill. For these econometric exercises, I focus on quarterly reports, which have been mandatory since 2008. The period of analysis runs from the first quarter of 2008 to the second quarter of 2014.

Table 7 shows the main results from these exercises. They show that firms belonging to the same subindustry as BP decreased their use of in-house lobbyists compared with a similar group of firms that was not affected by the oil spill. This is shown by the negative and significant coefficient measuring the interaction term. The decrease in the fraction of in-house reports is substantial. Given that the treated group made, on average, 19% of their reports in house before the spill, the point estimate represents a 26.3% decline in the fraction of reports made inside the firm. Regarding the fraction of in-house lobbyists, the average for the treated group before the spill was 24.7%. This represents a decrease of about 21% in the fraction of in-house lobbyists for firms in the oil and gas extraction industry.

5.2. BP and Skill Measures

The left panel of Figure 3 plots the *issue-specific* measures based on the fraction of high-skill witnesses for the treated and control groups. Two patterns emerge. First, as an additional confirmation of the validity of the control group, the time series for this group does not provide any suggestion of a shift around 2010. Second, the *issue specificity* of the lobbying activities for the treated group increased around the BP oil spill. The fraction of high-skill witnesses started increasing in the first semester of 2010. This series reached its peak in the first semester of 2012 and then, started decreasing. This represents an increase of about 69% in two years.

The event may have also modified the firm-specific skills of the industry. For instance, regulators and politicians reconsidered oversight of the industry, which led to a change in its regulation. The right panel of Figure 3 presents time series patterns in the number of regulating words for the treated and control groups. The figure shows that both control and treated groups increased the number of regulating words over this period. Remarkably, the firm-specific skills for the affected industry sharply decreased during the year of the spill.

As expected, Figure 3 also shows that the number of regulating words in the affected industry increased during the two years after the BP oil spill, a total

Table 7. Results of the BP Oil Spill Difference-in-Difference Estimation

	DV: Fraction of internal reports				DV: Fraction of internal lobbyists			
$T_i \cdot P_t$	−0.051** (0.024)	−0.052** (0.024)	−0.049*** (0.007)	−0.050*** (0.007)	−0.060** (0.028)	−0.062** (0.028)	−0.053*** (0.009)	−0.053*** (0.009)
T_i	0.078*** (0.020)	0.078*** (0.020)			0.071*** (0.023)	0.072*** (0.023)		
P_t	0.03*** (0.01)		0.015*** (0.003)		0.026** (0.013)		0.07 (0.05)	
Observations	3,729	3,729	3,729	3,729	3,184	3,184	3,184	3,184
Adjusted R^2	0.06	0.07	0.92	0.93	0.03	0.05	0.90	0.91
Fixed effects								
Quarter		Y		Y		Y		Y
Firm			Y	Y			Y	Y

Notes. An observation is a firm-quarter combination. The columns under fraction of internal reports provide the point estimates when the dependent variable is the fraction of reports made internally. The columns under fraction of external reports provide the point estimates when the dependent variable is the fraction of lobbyists working in house. Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the firm level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

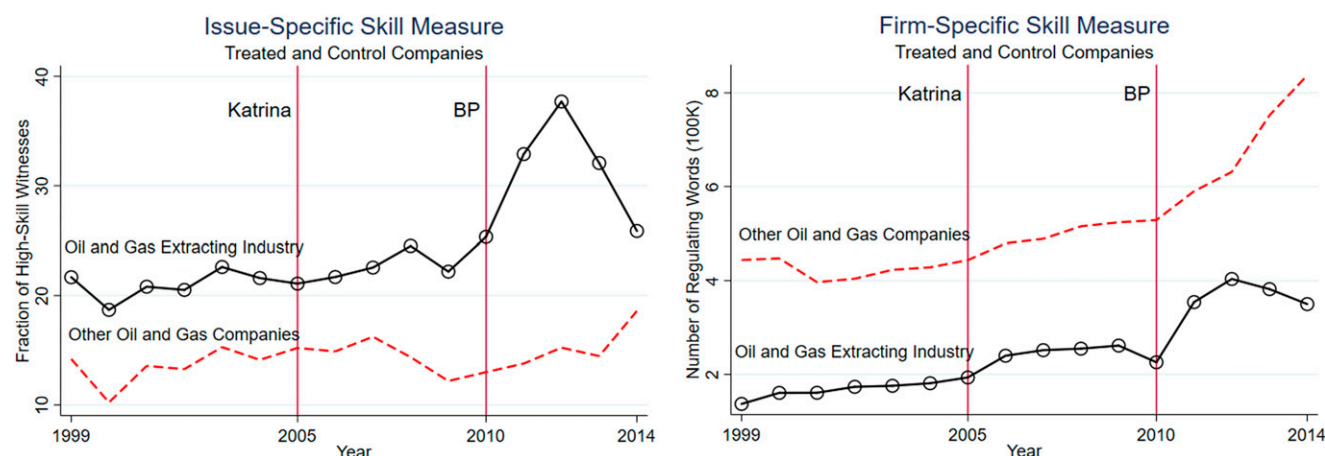
Significance at 5%; *significance at 1%.

increase of 54%. This may be a concern for interpretation of the results if the effect of the relative increase in the industry skills is close to or larger than the effect of a change in issue-specific skills for the affected group. However, this does not seem to be the case for two reasons. First, focusing on the treated group alone, the increase in regulating words is about 15 percentage points *lower* than the increase in the fraction of high-skill witnesses. Second, according to Table 3, the effect of a one-percentage point increase in the fraction of high-skill witnesses is about 10 times greater than the effect of a one-percentage point rise in the number of regulating words. Therefore, to claim that firm-specific skills drive the results around the BP

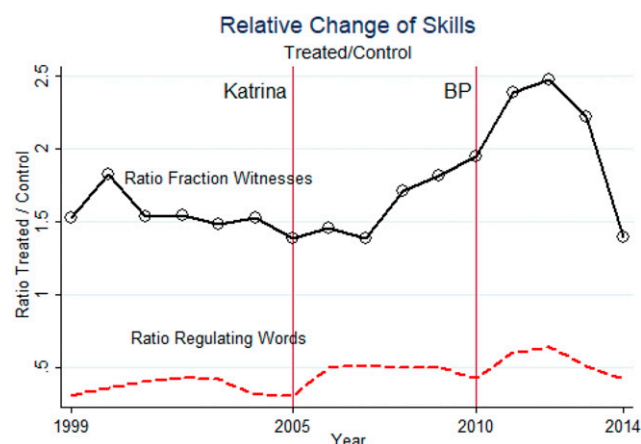
oil spill, we would need to see a change in such skills that is 10 times greater than the change in the fraction of high-skill witnesses. Empirical evidence is at odds with this possibility.

The analysis focuses on the raw data of the skill measures for both groups. Given that the econometric methodology uses difference-in-difference estimations, Figure 4 shows the time series of the skill measures for the treated group *relative* to the control group. For instance, the series of issue-specific skills shows the fraction of high-skill witnesses for the affected companies over this same fraction for the control group. Figure 4 shows, around the BP event, a marked increase in the relative measure for the issue-specific

Figure 3. (Color online) BP Oil Spill and Skill Measures



Notes. An observation is a group-year combination. The left panel shows the issue-specific variable measured by the fraction of high-knowledge witnesses. To construct this measure, for each semester, I consider all of the bills that these firms were lobbying. To get a group-level measure, I took the weighted average (by number of bills being lobbied) of the fraction of high-knowledge witnesses across all of the firms of the group. To get a yearly level measure, I take the average across semesters for the same year. The right panel shows the number of regulating words for these two groups. The vertical lines represent the years 2005 and 2010, the years in which Hurricane Katrina and the BP oil spill took place, respectively.

Figure 4. (Color online) Change of Skills for Affected Companies Relative to the Control Group

Notes. The figure presents the time series of the skill measures for the treated group relative to the control group. That is the evolution of the fraction of high-skill witnesses and number of regulating words for the affected companies over the skill measures for the control group.

skills and an insignificant increase in the measure of firm-specific skills.³⁰ That is, even if the firm-specific skills of the oil and gas extraction industry rose after the oil spill event, this was *also* accompanied by a similar increase in the regulation of firms in the control group. This suggests that the changes in the regulating words of the sector manifest a macro-level evolution in the regulation of the whole oil sector of a type that is independent of the BP event.

All in all, Figures 3 and 4 suggest that the effect of the BP oil spill can be interpreted as a disproportionate increase in issue-specific skills but not in the number of regulating words of the industry. Intuitively, it is natural to expect that issue-specific skills react more strongly to shocks than firm- or industry-specific skill measures based on the intensity of regulation.

For instance, changing the type of witnesses in congressional hearings is much faster and easier than making economic regulations by, for example, converting a bill into law.

In the following subsection, I present empirical estimations that support the validity of the baseline results.

5.3. Robustness Exercises

The oil spill may have increased the number of problems faced by the treated group. That is, it may have sparked a need to scale up advocacy activities. Given the in-house staff's time constraint, an increase in the number of problems may have automatically increased the need for service providers. Thus, the use of external service providers may be explained by the time constraint of in-house staff and not by a change in *issue specificity*. A way of testing the validity of this story is to see whether there were changes in the absolute number of in-house lobbyists. Table 8 provides evidence that firms in the treated group decreased the use of internal staff and increased the use of lobbyists for hire. An additional way to control for the change in the number of problems that the firms faced is to run the same econometric specifications as the baseline estimation but including a measure of the firm's workload as an additional control variable. Table 7 in the e-companion presents coefficient estimates after the total number of SOPR reports is included as an additional explanatory variable. The main results presented in the baseline estimations still hold. These tables are consistent with a story in which the primary motivation for outsourcing is to access specialized skills rather than the expected pattern of contracting intended to smooth the firm's workload.

An alternative concern is that there may have been negative spillovers to other firms in the oil and

Table 8. Evidence of the Absolute Change on the Number of Lobbyists

	DV: No. of internal lobbyists				DV: No. of external lobbyists			
$T_i \cdot P_t$	-0.174** (0.072)	-0.188** (0.078)	-0.169** (0.069)	-0.1925** (0.079)	0.360** (0.145)	1.405*** (0.199)	0.382** (0.159)	1.448*** (0.198)
T_i	-0.409*** (0.148)		-0.404*** (0.148)		2.516*** (0.338)		2.517*** (0.338)	
P_t	-0.0231 (0.0837)	-0.0950 (0.122)			-0.132 (0.194)	-0.394*** (0.102)		
Observations	3,729	3,729	3,729	3,729	3,729	3,729	3,729	3,729
Adjusted R^2	0.361	0.841	0.363	0.843	0.055	0.826	0.058	0.829
Fixed effects								
Quarter			Y	Y			Y	Y
Firm		Y		Y		Y		Y

Notes. An observation is a firm-quarter combination. The dependent variable is the total number of internal lobbyists or the total number of external lobbyists. Observations are the numbers of observations used to run the econometric estimations. Standard errors are clustered at the firm level. The lower section represents the fixed effects used. Y denotes that a specific set of controls has been used.

Significance at 5%; *significance at 1%.

gas industry. Although Figures 2 and 3 do not seem to show any change for the control group around the event, in Table 8 in the e-companion, I present additional econometric estimations using *three* different control groups and show that the baseline results hold.³¹ This implies that the selection of control groups does not explain the results.

An additional concern is that the variation in the interaction variable is not at the firm level but at the industry level. To control for this type of autocorrelation, I show that the main results are robust when I two-way cluster the standard errors (*industry* · *quarter* and *firm*), or such as in Barrage et al. (2020), I aggregate the data at the firm-period level (two periods: before and after the spill). Tables 9 and 10 in the e-companion show that the results from these two sets of exercises are qualitatively similar to the baseline estimation.

Previously, I argue that the change in skill requirements brought by the oil spill may explain the change in the outsourcing decision around the BP case. However, other simultaneous changes can confound this relationship. For instance, oil spills increase the demand for policy change for which firms may find it optimal to outsource the lobbying service (i.e., external lobbyists have more connections with policy makers). If this is the case, oil spills should increase the use of external lobbyists. To advance in this direction, I run estimations of variables measuring the damage caused by the major oil spills that occurred in the period 2008–2014 (18 in total) on integration measures.

Tables 11 and 12 in the e-companion show that oil spills (measured with a dummy variable that takes the value of one when there is an oil spill in the semester and zero otherwise; number of tonnes spilled or number of oil spills) do not statistically explain integration patterns. This suggests that changes in skill needs rather than the intrinsic nature of oil spill events that might have brought policy change explain the lobbying labor boundaries for firms in the oil and gas industry.

A remaining concern is that integration may not be modified by any major oil spill but only by the largest ones having certain characteristics. To explore this idea, we seek to find an event that shares relevant resemblance to the BP event but differs in its effects on the issue-specific skill requirements. Finding such an event, although challenging, would prove useful for testing various implications. For instance, the results presented in the baseline estimations are consistent with a theory in which clients hire external lobbyists to bring the legitimacy that in-house lobbyists may have lost after the event.³²

In the following section, I argue that the oil spills provoked by Hurricane Katrina share several similar

characteristics with the BP event but that, importantly, the former was not accompanied by a significant increase in the fraction of high-skill witnesses. Then, I show that lobbying integration measures were not modified around Hurricane Katrina.

5.4. Distinguishing Skill Needs from Other Explanations: Katrina Event

Hurricane Katrina, a category 5 hurricane, was one of most devastating hurricanes in the history of the United States.³³ The hurricane took place over the last week of August 2005 and affected U.S. oil-producing regions. The Department of Environmental Quality estimated that 44 oil spills took place, and taking them all together, Hurricane Katrina provoked one of the worst oil spills in the history of the United States.³⁴

Although this powerful hurricane was undeniably responsible for the spills, both their number and magnitude could have been reduced by better safety practices in the oil industry. In the e-companion of this paper, I present examples that show that some of these oil spills were owing to the negligence, poor safety standards, and lack of foresight from the oil companies. This lack of adequate care from oil companies, just as in the case of the BP oil spill, caused a stigma to be attached to the oil and gas extraction industry. This stigma was manifested in different ways. For instance, the general public reacted with legal action, and congressmen proposed policies and bills to negatively affect the industry.

In order to use Hurricane Katrina as a placebo test, it is important to verify that the periods surrounding the events of Hurricane Katrina and the BP oil spill present similarities in the political environment because failure in this assumption might explain differences in the effects that these events have on integration patterns. I use information from the *Congressional Bills Project* to show that the political environment around Hurricane Katrina is statistically indistinguishable from that of the BP oil spill.³⁵ Table 13 in the e-companion of this paper compares characteristics of the bills in the third quarter of 2005 when Hurricane Katrina took place with the quarter in which the BP oil spill occurred. The table shows that there are no statistically significant differences in the characteristics of the bills related to the oil and gas industry between Hurricane Katrina and BP periods.

Another relevant variable, the number of regulating words of the affected industry, evolved similarly around the Hurricane Katrina and BP oil spills. As shown in Figure 3, firm-specific skills decreased during the year of the BP event (2010), and then, they increased in the following two years: first sharply and then, gradually. The regulating words behaved similarly around 2005. The series did not significantly increase from 2004 to 2005 (only by 1.75%).

Then, in 2006, there was a huge increase followed by a more gradual rise in 2007.

Other than these similarities, important for identification, Figures 3 and 4 show that the fraction of high-skill witnesses relevant to the oil and gas companies did not change significantly around Hurricane Katrina. This fraction was 23.59 in 2004, and it slightly decreases to 4.4%. However, the proportion of high-skill witnesses in 2009 was 22.18, and it rose by 69.93%.

Given the arguments, Hurricane Katrina is a suitable candidate to distinguish the stories of legitimacy from the story proposed by this article: skill acquisition. The legitimacy story predicts that vertical integration should increase even when the fraction of high-skill witnesses does not change. To test this idea, I regress different integration measures, and I include firm fixed effects as well as semester fixed effects to control for unobserved heterogeneity. I restrict these estimations to the period 1999–2007, and as before, I define the treated group as the firms in the oil and gas extracting industry, excluding those directly affected by the oil spill.³⁶

Table 14 in the e-companion shows that, for any of the integration measures and for any of the control groups used, there is no evidence of a statistical change in lobbying integration measures after Hurricane Katrina. Therefore, it seems that just the event of a large oil spill, in the absence of a change in issue-skill measures, does not significantly modify integration patterns in the U.S. lobbying industry.

This exercise does not disprove the validity of the need to bring legitimacy in the advocacy process. This can be the case because it may be that, even though firms face stigma after an oil spill, in-house lobbyists do not lose legitimacy because, like external lobbyists, they too have a reputation to protect. Therefore, firms would not outsource lobbyists to bring legitimacy because the oil spills do not change their relative legitimacy needs.

Based on the evidence presented in this section, I interpret the increase in lobbying outsourcing around the 2010 BP oil spill as the response of the firms affected to an increase in the issue-skill requirements of advocacy activities.

6. Discussion and Conclusions

6.1. Discussion of the Literature

There is a large set of theories explaining firm boundaries. At least four economic theories deserve mention here. In this section, I argue that *some* of them may not be a natural starting point to interpret the context of this article (property rights theory and rent-seeking theory (RST)). However, even if I study natural interpretations of all *four* of these theories, (1) they require other kinds of detailed data that unfortunately

this article does not have, and (2) empirical evidence provided in this article does not support some of their theoretical implications.

6.1.1. Property Rights Model. This theory focuses on neither contractible nor alienable investments to understand firm boundaries.³⁷ Although this literature emphasizes the ownership of nonhuman assets as a tool to exercise power in an incomplete contracts world as suggested by Dube and Kaplan (2007), the theory does not have clear implications when such assets are irrelevant as in the case of outsourcing service industry workers. Clearly, the difference is that the law does not provide control rights over human beings: buying a machine is totally different to hiring someone in-house because the employee can always quit.

An interpretation of the property rights model (PRT) is provided by Whinston (2001, 2003). He considers a bilateral trade setting in which a seller (lobbyist) uses an asset for the production of a product required by a buyer (firm). He argues that, under the assumption that the seller's investment creates knowhow for herself, a larger importance of specific knowhow leads to more outsourcing. To investigate the empirical validity of PRT, he uses the work of Monteverde and Teece (1982), who find a positive empirical relationship between specific knowhow and integration. As a consequence, Whinston (2003, pp. 16–17) concludes that the empirical results of Monteverde and Teece (1982) contradict this interpretation of the PRT.³⁸ One can interpret the firm-lobbyists relationship as the one proposed by Whinston (2003). Firms need to carry out tasks, and lobbyists exert noncontractible efforts to develop knowhow that is specific to the relationship. An increase in the importance of this knowhow (for instance, because of a rise in the firm-specific needs) will increase outsourcing according to the PRT. Tables 3–5 show that, when the firms have larger firm-specific needs, they are more likely to integrate rather than outsource. Therefore, the empirical setting of this article contradicts this interpretation of the PRT.³⁹

6.1.2. Multitasking Theory. One of the key predictions of Holmstrom and Milgrom (1991) is that, when there are two tasks that differ in the cost of measuring performance, firms bring in house the tasks that are harder to measure. An interpretation of this theory in the advocacy context is that lobbyists can exert effort in two tasks: acquiring greater knowledge of the subject or executing the duties in a timely manner. The first task is a noncontractible job, whereas the second is contractible. As is standard in the literature, assume that contracts of external lobbyists differ from those of in-house advocates because the former

has high-powered incentives and the latter has low-powered incentives. Multitasking theory implies that, given this arrangement of incentives, external lobbyists would only put effort into the observable dimension (speed) but not into acquiring greater knowledge. Given the low-powered incentives of in-house lobbyists, they are more likely to balance effort between these two tasks, and therefore, they would be slower than external lobbyists but would invest in acquiring more in-depth knowledge of the subject. This implies that tasks that require greater depth (for instance, those that are studied with a larger fraction of high-skill witnesses) would be performed by in-house rather than external lobbyists. Tables 3–5 contradict the predicted patterns of this interpretation of the theory. To find a definite answer on the explaining power of this theory in the advocacy context, we need to have more information on the type of contracts (high- versus low-powered incentives) that lobbyists have. Unfortunately, the data studied in this article do not contain this type of information.

6.1.3. Adaptation Theory. This literature studies how integration facilitates adaptive decision making in uncertain environments.⁴⁰ Adaptation theory is relevant in my context because the complexity of the political process prevents lobbyists and firms from writing contracts that specify how advocates act when contingencies that arise need to be solved in a timely way. As a consequence, ex post adaptation becomes necessary.⁴¹ To investigate the relationship between ex post adaptation needs and integration, I study the number of amendments (*defined as any proposed change to a bill*) reported for each bill for the period 1999–2014 from Congress.gov.⁴² As with the bill-level measures, I use the average number of amendments at the committee-semester level and match it with lobbying reports. This average measure provides an estimate of the probability that firms will be subject to ex post adjustments. Adaptation theory predicts that firms that face a larger probability of ex post adjustment needs would be more prone to bring lobbyists in house. Table 15 in the e-companion presents fixed effect estimations of the average number of amendments on different measures of vertical integration. The table shows that neither the fraction of in-house lobbyists or reports nor the number of lobbyists are significantly correlated with the measure of amendments.

A way to reconcile these findings with adaptation theory is that integration is the preferred mode of transaction when external advocates take actions that negatively affect firms after contingencies arise. However, this does not seem to be the case because anecdotal evidence suggests that external lobbyists

work for firms that, although they belong to different industries, have common goals.⁴³ This implies that external lobbyists cannot act in favor of one firm rather than another when uncertainty is revealed, and therefore, the goal-alignment advantage that integration offers is less important.

6.1.4. Rent Seeking Theory (RST). This literature predicts that integration can end inefficient haggling over *appropriable quasirents*.⁴⁴ However, as Gibbons (2005) states, integration can only modify certain types of haggling. If the haggling is done with the manipulation of inalienable assets, such as the skills that this article focuses on, then integration cannot modify rent-seeking behavior. That is, rent-seeking models that predict integration as a solution to haggling can only work for holdup situations that utilize alienable instruments. Therefore, it seems that this theory, although empirically confirmed in many contexts, may not be a natural starting point to study the empirical patterns of this article.

A deeper issue arises in regard to this theory when we focus on the firm-specific needs dimension that this article studies. One of the predictions from the RST is that integration is more likely when the *firm-specific* need of the transaction increases. The idea is that the more specific the transaction is, the more likely it is that the firm will face a holdup problem; thus, to avoid it, the activity should be conducted internally.⁴⁵ However, as Klein (1988) points out, this literature does not clarify how this mitigation occurs in the case of inalienable assets, such as the skills that this article focuses on.⁴⁶

We can, however, try to test the validity of *one* interpretation of the theory for the advocacy industry. Firms that face more frequent changes in the environment (a measure of uncertainty) will face more haggling. Therefore, if we assume that integration mitigates holdup problems, firms that face more of these changes will be more likely to hire in-house lobbyists. The number of amendments to the bills may measure this uncertainty, and therefore, a potential prediction of this theory would be that firms that face more amendments use more in-house lobbyists. Table 15 in the e-companion does not provide evidence that this is the case.

6.2. Conclusions

Abstracting from the current application, the analysis underscores the potential of nonincentive integration ideas to explain how firms use knowledge workers. I believe that this article is a small step toward a better understanding of the integration decision with service workers. However, this article faces the limitations intrinsic to any study trying to understand abstract concepts,

such as knowledge and skills. I collected a new data set in an attempt to meet this challenge, but limitations remain.

Promising research avenues include exploration of the insights developed here to understand the causes and consequences of advocacy activities as well as the consequences of the integration decision in this or other industries.⁴⁷ The theoretical framework studied in this article is static, and therefore, future work should explore how the insights gained from this article can be enriched when we study a dynamic setting and allow for the possibility that firms' needs and lobbyists' skills evolve over time.⁴⁸

The theoretical framework and anecdotal evidence provided here point toward the validity of firm-specific skills as an important force for explaining integration patterns. Data availability constrained me from conducting empirical exercises with firm-level variation. Future studies should aim to complement the exercises proposed here by filling this gap.

At least since Williamson (1985), it has been predicted that, in order to save on the negotiation costs required in each transaction, firms are more likely to bring transactions in house when they occur frequently. The theoretical framework in this article predicts the same relationship between frequency and integration, but this is owing to a desire not to save on negotiation costs, but simply to receive increasing returns from the use of skills of in-house personnel. The current data do not allow me to differentiate between these two theories, and future work on this distinction is warranted.

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Endnotes

¹ Berlingieri (2014) shows that the professional business services industry (NAICS codes 54, 55, and 56) exerts the biggest influence on the rest of the economy in terms of forward linkage measures. He also shows that this industry has the largest change in input-output linkages, an increase close to four times in 60 years, and that it is the subindustry with the greatest growth.

² Two early attempts to study the integration question when physical assets are irrelevant are Anderson and Schmittlein (1984) and Anderson (1985). These two papers use a different approach to the one taken here because they study the integration under the lens of transaction cost economies.

³ See, for instance, Grossman and Helpman (2002), Ansolabehere et al. (2003), Antras and Helpman (2004), de Figueiredo and King (2004),

de Figueiredo and Silverman (2006), Bombardini (2008), Richter et al. (2009), Bombardini and Trebbi (2012), Kerr et al. (2014), Hirsch and Montagnes (2015), de Figueiredo and Richter (2014), Kang (2016), and Lambert (2019).

⁴ For a review on the empirics of vertical integration, see Lafontaine and Slade (2007). For the theoretical literature, see, for instance, the property rights model and the multitasking, adaptation, and rent-seeking theories. All of these theories are surveyed and compared with the approach taken in this article in Section 6.1.

⁵ The theory studied here does not focus on the incentives brought by long-term relationships. In Section 2 of the e-companion, I discuss the implications of this theory to my empirical context.

⁶ The LDA defines a lobbyist as a person spending 20% or more of her time engaged in lobbying activities.

⁷ For the sake of a better comparison, for most of the estimations, I focus on semester-level time variations.

⁸ Although lobbying reports contain information on expenditures, I exclude them from my analysis because in-house reports include nondisaggregated expenditures that are not directly related to advocacy activities (e.g., office space).

⁹ Even narrow topics, such as advertising, span several firms. There were about 51 companies actively lobbying for this topic in 2014.

¹⁰ All of the bills are allocated to at least one committee, and most of the bills go to only one. I focus here on standing committees, which are permanent bodies with specific responsibilities. In the *House of Representatives*, bills' referral is controlled by the *Speaker* following the advice of the *House Parliamentarian*. In the *Senate*, it is managed mainly by the *Senate Parliamentarian* on behalf of the presiding *Officer of the Senate*. *Parliamentarians* are nonpartisan officials that provide technical assistance and expertise on the legislative procedure of the Congress. They serve for several years; indeed, there have been only five parliamentarians in each house since 1928.

¹¹ For more information, see Sachs (1999), Sullivan (2007), and Heitshusen (2015).

¹² In Section 4 in the e-companion, I discuss this classification in more detail and provide examples of each of these three categories. I also present a validation exercise using data from the *O*NET* program. To minimize any noise caused by this classification, the empirical results use the fraction of high-knowledge witnesses over all of the classified witnesses: that is, I exclude the unclassified occupations. Alternative estimations, when the denominator includes the category of unclassified occupations, do not qualitatively modify the results presented here.

¹³ This is similar in spirit to the work by Azoulay (2004), in which the proportion of academic researchers (over the total number of participants in clinical studies) captures the relative importance of knowledge-intensive activities.

¹⁴ Proxying regulation with this type of measure is not new. Mulligan and Shleifer (2005) and Coffey et al. (2012) proxy the extent of regulation of the whole economy with the number of pages and size of digital versions of regulations, respectively. The innovation of RegData is to include regulation measures at the industry and not at the economy-wide level.

¹⁵ These rules come from two main resources: congressional bills that become laws and regulations made by federal agencies. The code is divided into 50 titles representing broad subject areas in federal regulations, such as agriculture, energy, banking, and public health.

¹⁶ This stability does not mean that the regulation for a specific industry does not change over time. This is manifested for instance, in Table 2, which shows substantial within-industry variation.

¹⁷ For examples of the relevance of union status, see Abraham and Taylor (1996), and for examples of geographical issues, see Autor (2003).

¹⁸For the effect of the financial crisis on integration patterns, see Knudsen and Foss (2014), and for economy-wide related time-varying patterns, see Abraham and Taylor (1996).

¹⁹To clarify this process, let us assume that a lobbying report in period t has information on two bills: bill 1 and bill 2. In this period, bill 1 has been assigned to committee 1, and bill 2 has been assigned to committee 2. Assume that the fractions of high-skill witnesses in the congressional hearings for committees 1 and 2 in period t are X and Y , respectively. This lobbying report will, therefore, have an *issue specificity* measure equal to the average between X and Y . When the reports have more bills than committees, I use a weighted average as explained in footnote 20.

²⁰Let b_{cjt} be the total number of bills being lobbied for firm j in period t sent to committee c . Let f_{ct} be the frequency (total number of bills) received by committee c in period t . The frequency measure for firm j in period t is $\sum_c f_{ct}(\frac{b_{cjt}}{\sum_c b_{cjt}})$.

²¹For the issue of inertia in lobbying activities, see Kerr et al. (2014).

²²For instance, for the issue-specific skills with dependent variable VI and firm and semester fixed effects, the coefficient is -0.0271 according to Table 3. This coefficient is -0.0250 when I include the frequency variable in Table 4.

²³For a recent discussion on the issue of generalists and specialists and their relationship with relational capital in the lobbying market, see Byun et al. (2018). My paper, contrary to that paper, focuses on how the classification of generalists/specialists is related to lobbyists' integration status.

²⁴Notice that the definitions of specialists and generalists are not exhaustive. There may be lobbyists who do not match any of these classifications.

²⁵I cluster standard errors at the lobbyist level.

²⁶Scientists estimated that the total amount of released oil was close to 4.9 million barrels according to the United States Coast Guard report. The report can be found at <https://www.gpo.gov/fdsys/pkg/GPO-OILCOMMISSION/pdf/GPO-OILCOMMISSION.pdf>.

²⁷Barrage et al. (2020) show that stations selling BP-branded fuels experienced a damaging effect on both prices and sales. They interpret this finding as a shift in demand away from BP as a way to punish the company's bad practices.

²⁸Furthermore, several bills were introduced after the event with the aim of making the oil and gas extraction business more difficult. For instance, bill H.R. 5436 proposed prohibiting the issue of permits for any deep-water drilling in the Gulf of Mexico; bill H.R. 5222 proposed suspending exploration and production activities in the outer continental shelf until the investigation of the BP oil spill had concluded.

²⁹For more information, see Reed and Krauss (2010).

³⁰Although there was an increase in the fraction of high-skill witnesses in the years before the BP oil spill, there was no differential effect on the integration patterns for the previous quarters between the treated and control groups. This is confirmed by the leads and lags exercise shown in Figure 3 of the e-companion.

³¹The industries for these alternative control groups are *retail sales*, *real estate*, and *casino and gambling*. These groups were selected according to the preoil spill similarity of trends in the fraction of internal reports.

³²For a discussion on the issue of lobbying outsourcing and legitimacy needs, see Jia (2018, p. 321).

³³For more information, see the website of the National Oceanic and Atmospheric Administration at <https://www.nhc.noaa.gov/outreach/history/>.

³⁴For more information, see Llanos (2005).

³⁵For more information, see the website of the Congressional Bills Project at <http://www.congressionalbills.org/>.

³⁶The oil spills came from Bass Enterprises Oil Spill, Shell, Murphy Oil USA, Chevron, Venice Energy Services Company, and Sundown Energy according to Llanos (2005).

³⁷See, for example, Grossman and Hart (1986), Hart and Moore (1990), and Hart (1995).

³⁸Whinston (2003, p. 17) claims: "Thus the predictions of this property rights model contradict M&T's findings regarding the effects of increased know-how."

³⁹This interpretation of the PRT should not make the reader conclude that this theory cannot explain some of the patterns studied here. In order to clarify the validity of the PRT for the empirical context of advocacy, we would need to identify some exogenous "variation whose effect on marginal returns to investments in various ownership structures seems very clear a priori" (Whinston 2003, pp. 20–21). However, the data used in this study are not appropriate for a definite test of this theory. For instance, it is not clear how to measure non-contractible investments of both firm and lobbyists and how to estimate the disagreement payoffs and their relationship with investments.

⁴⁰The main insights from this literature can be traced back to Simon (1951), Williamson (1971, 1973, 1975, 1991), and Klein and Murphy (1988). A recent survey of the main results of the transaction cost economies that includes both adaptation and rent-seeking theories can be found in Tadelis and Williamson (2012).

⁴¹Important examples of such contingencies are amendments to the text of bills. Firms and advocates meet to decide on lobbying strategies for political bills. However, these are studied in committees, and they are subject to changes that may modify the returns of the policy to the firm. Such changes are usually subject to approval within short periods, and therefore, the lobbyists' strategies on how to tackle new versions of the bills need to be adapted.

⁴²For more information, see the website of [Congress.gov](https://www.congress.gov), the official website for U.S. federal legislative information.

⁴³A typical example of lobbyists who represent a set of firms with common goals is those aiming to decrease corporate taxes or tariffs on input imports.

⁴⁴The main insights from this literature can be traced back to Williamson (1971, 1979, 1985) and Klein et al. (1978). An early empirical example of this literature is Anderson (1988).

⁴⁵See, for instance, Klein et al. (1978), Monteverde and Teece (1982), and Anderson and Schmittlein (1984).

⁴⁶An additional issue that we need to solve to apply this theory to the integration of human assets is that, if the holdup problem depends on firm-specific skills and these skills increase under integration as the empirical evidence suggests, then the holdup is not solved but aggravated under integration.

⁴⁷For examples of recent articles studying variables explaining the lobbying decision, see Bombardini (2008), Kerr et al. (2014), and Hegde and Sampat (2015). For a recent example on the consequences of lobbying, see Lambert (2019).

⁴⁸See, for instance, the insights developed by Stigler (1951) and Langlois (1992, 2006).

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